House Price Prediction

A full-stack Machine Learning web application that predicts house prices based on **area (sq ft)** and **number of rooms**.

Project Overview

This project demonstrates a complete machine-learning workflow:

1. Load and prepare a house-price dataset.
2. Train a Linear Regression model.
3. Save the trained model for prediction.
4. Provide a Flask REST API backend.
5. Provide a Streamlit frontend for user interaction.
6. Generate predictions from the trained model.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Flask
- Streamlit
- Joblib

Dataset

The project uses a house-price dataset containing the following main features:

- `area` — house area in square feet
- `rooms` — number of rooms
- `price` — house price (target variable)

Dataset file:

`data/house_price.csv`

Project Structure

```text
house_price_project/
│
├── backend/
│   └── app.py
│
├── data/
│   └── house_price.csv
│
├── frontend/
│   └── ui.py
│
├── model/
│   └── model.pkl
│
├── report_images/
│
├── train_model.py
├── requirements.txt
├── README.md
└── app.py
```

> File names may be adjusted according to the final project folder.

Machine Learning Model

The project uses **Linear Regression** to predict house prices.

Input Features

- Area (sq ft)

- Number of rooms

Output

- Predicted house price

The trained model is saved as:

`model/model.pkl`

Installation

Clone/download the project and open a terminal in the project directory.

Install the required packages:

```bash
pip install -r requirements.txt
```

Train the Model

Run:

```bash
python train_model.py
```

This trains the Linear Regression model and creates the saved model file.

Run the Flask Backend

Start the backend with:

```bash
python backend/app.py
```

```

The Flask API can then be used to send house details and receive a prediction.

## Run the Streamlit Frontend

Start the user interface with:

```bash
streamlit run frontend/ui.py
```

A browser window will open with the house-price prediction interface.

Example

Enter:

- Area: `1200`
- Rooms: `3`

The application sends these values to the trained model and displays the predicted house price.

Project Objective

The objective of this project is to build a simple end-to-end machine-learning application that connects:

Dataset → Model Training → Flask API → Streamlit UI → House Price Prediction

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License
```

This project is created for educational and internship/project-submission purposes.